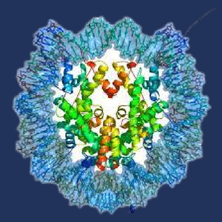

in-silico Evaluation of Dietary Fish Fatty Acids as Potential Atheroprotective Therapeutics

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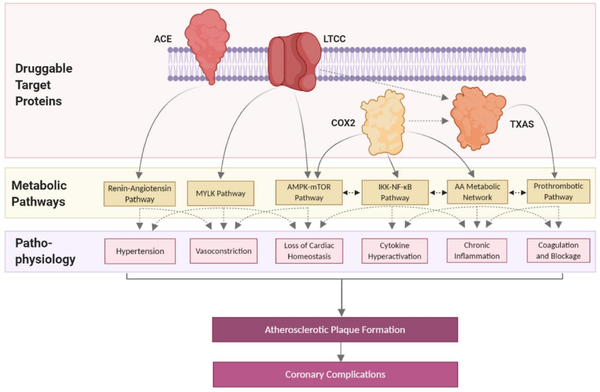
Background of Study

Atherosclerosis is a complex vasoconstrictory condition that is associated with cardiovascular, pulmonary and immunomodulatory dysfunctions. Hypertension, chronic inflammation and homeostatic dysregulation lead to lipid plaque formation in arteries and eventually result in cardiac infractions and tissue damage.

Current therapeutic approaches target different metabolic pathways and druggable proteins for direct or indirect atheroprotective effects. These targets majorly comprise of pro-inflammatory enzymes like Cyclooxygenase 2 (COX2) and Thromboxane A2 Synthase (TXAS); another mode of action is by blocking Angiotensin I Converting Enzyme (ACE) and L-Type Ca^{2+} Channel protein (LTCC) which promote arterial constriction. However, these drugs are selected based on single drug-single enzyme mechanisms while the target enzymes themselves are linked via complex feedback loops and intercommunications– leading to unexpected and devastating side effects in patients.



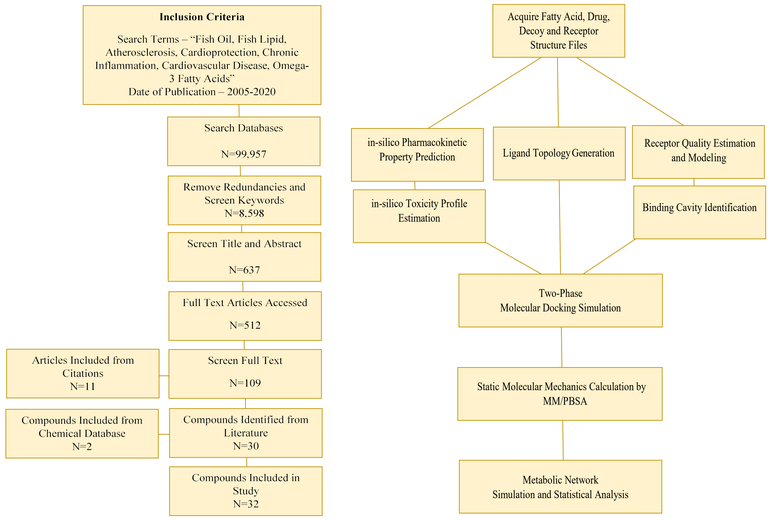
Fish fatty acid supplements have been touted for their cardioprotective roles as safer natural alternatives. Few yet substantial pre-clinical experiments have also supported claims of anti-oxidant, anti-inflammatory and endothelial tissue maintenance properties of these fatty acids. However, there have not been significant attempts at elucidating the exact molecular mechanisms and receptor interactions that could help explain these bioactivities.

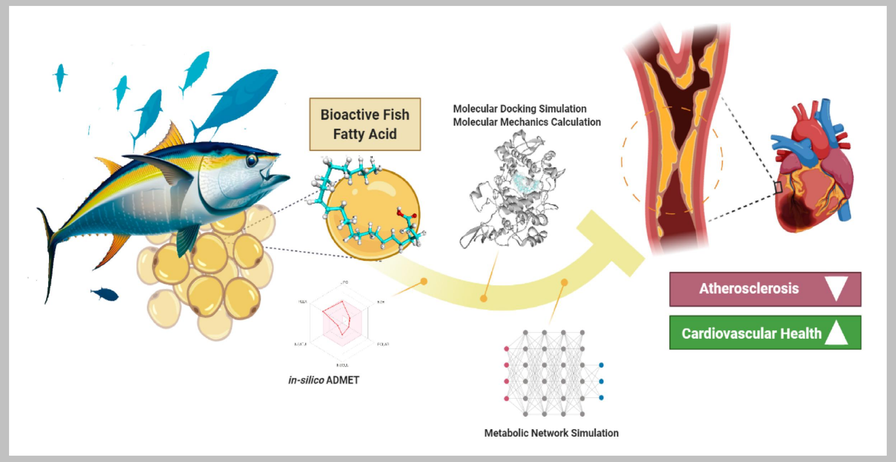
Here we have constructed a library of bioactive lipids derived from dietary fish sources through systematic literature review and subjected them to standard computational screening and simulation protocols. The simulations provide an initial prediction of efficacy, safety and bioavailability of the studied fatty acids and elucidate specific single and multi-enzyme interaction behaviors with atherosclerosis associated target pathways. The findings of this study act as a seminal foundation for fish lipid biologics in targeted anti-inflammatory and atheroprotective therapy.

Objectives

- Provide molecular insights into the atheroprotective activities of reported bioactive fish fatty acids through computational investigation of multiple metabolic pathway modulation potentials
- Predict potent and safe LTCC, ACE, COX2 and TXAS inhibiting atheroprotective drug-like compounds from bioactive fish fatty acids

Experimental Section





Results

	Dietary Fish Lipids	Pharmacokinetic Properties: GI Absorption	BBB Permeant	Drug-Likeness: Bioavailability Score	Drug-Likeness	Toxicity Profiles: Cardiotoxicity	Hepatotoxicity	Carcinogenicity	Allergenicity
A	Caprilic Acid	High	Yes	0.85	Yes	No	No	Probable	No
	Lauric acid	High	Yes	0.85	Yes	No	No	Probable	No
	Myristic Acid	High	No	0.56	Yes	No	No	No	No
	Myristoleic Acid	High	Yes	0.85	Yes	No	No	No	No
	Pentadecanoic Acid	High	Yes	0.85	Yes	No	No	No	No
	Palmitic Acid	High	Yes	0.85	Yes	No	No	No	Probable
	Palmitoleic Acid	High	Yes	0.85	Yes	No	No	No	No
	Hexadecadienoic Acid	High	Yes	0.85	Yes	No	No	No	No
	Hexadecatrienoic Acid	High	Yes	0.85	Yes	No	No	No	No
	Hexadecatetraenoic Acid	High	Yes	0.85	Yes	No	No	No	No
	Heptadecanoic Acid	High	Yes	0.85	Yes	No	No	No	No
	Stearic Acid	High	No	0.85	Yes	No	No	Probable	No
	Oleic Acid	High	No	0.85	Yes	No	No	No	Probable
	Vaccenic Acid	High	No	0.85	Yes	No	No	No	No
	Linoleic Acid	High	Yes	0.85	Yes	No	No	No	No
	Linolenic Acid	High	Yes	0.85	Yes	No	No	No	No
	Steridonic Acid	High	Yes	0.85	Yes	No	No	No	No
	Arachidonic Acid	Low	No	0.85	Yes	No	No	No	No
	Eicosanoic Acid	Low	No	0.85	Yes	No	No	No	Probable
	Gadoleic Acid	Low	No	0.85	Yes	No	No	No	No
	Eicosedienoic Acid	High	Yes	0.85	Yes	No	No	No	Probable
	Eicosetrienoic Acid	High	No	0.85	Yes	No	No	No	Probable
	Eicosapentaenoic Acid	High	No	0.85	Yes	No	No	No	Probable
	Heneicosanoic Acid	Low	No	0.85	Yes	No	No	No	No
	Behenic Acid	Low	No	0.85	Yes	No	No	No	No
	Docosapentaenoic Acid	Low	No	0.85	Yes	No	No	No	No
	Docosahexaenoic Acid	High	No	0.85	Yes	No	No	No	No
	Tricosanoic Acid	Low	No	0.85	Yes	No	No	No	No
	Lignoseric Acid	Low	No	0.85	Yes	No	No	No	No
	Octadecatetraenoic Acid	High	Yes	0.85	Yes	No	No	No	No
	Erucic Acid	Low	No	0.85	Yes	No	No	No	Probable
	Nervonic Acid	Low	No	0.85	Yes	No	No	No	No

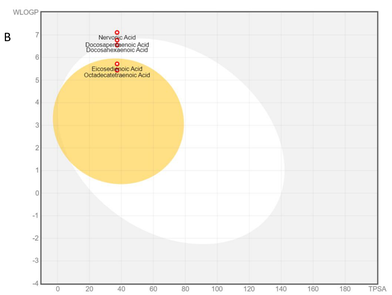


Fig 1: in-silico ADMET properties of dietary fish lipids A) predicted pharmacokinetics, drug-likeness and toxicity profile values of fish lipids; B) "Boiled egg" diagram of top performing fatty acid molecules showing highly GI absorbed (white) drug like molecules and BBB permeants (yolk).

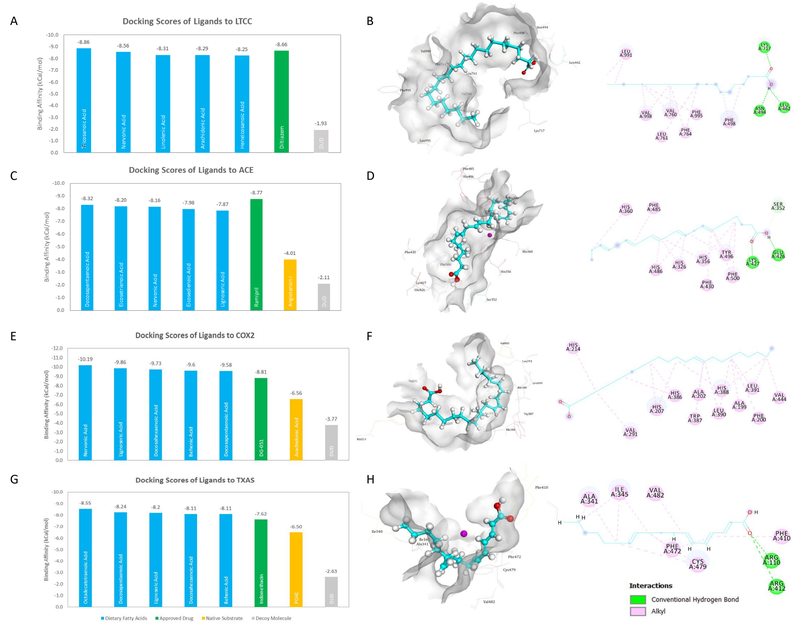


Fig 2: Summarized docking results of dietary fish fatty acids to target proteins. Docking scores of top 5 performing fatty acids, positive controls and decoy molecules against A)LTCC C) ACE E)COX2 and G) TXAS; Best docked 3D conformations and 2D interaction plots of top performing fatty acids surrounded by interacting amino acid residues (lines) and cofactors (pink discs) for B) Tricosanoic acid with LTCC D) Docosapentaenoic acid with ACE F) Nervonic Acid with COX2 and H) Octadecatetraenoic acid with TXAS complexes.

Results (Cont.)

Table 1: Molecular Mechanics Calculation of top docked fatty acid and drug molecules to target receptors

Energy Component (kCal/mol)	LTCC: Tricosanoic Acid	LTCC: Diltiazem	ACE: Docasopentaenoic Acid	ACE: Ramipril	COX2: Nervonic Acid	COX2: DG-051	TXAS: Octadecatetraenoic Acid	TXAS: Indomethacin
ΔE_{VDW}	-63.70±2.64	-83.82±0.31	-101.80±2.45	-107.98±0.31	-64.51±5.89	-76.51±8.26	-91.88±4.41	-95.82±9.28
ΔE_{EL}	-87.54±4.65	-60.30±5.11	-93.68±6.68	-36.30±9.05	-73.85±3.22	-47.95±3.72	-69.30±4.33	-18.30±9.48
ΔG_{POL}	117.24±5.40	111.82±6.79	181.68±1.64	156.76±9.12	160.49±3.52	122.07±6.36	154.79±4.52	137.47±3.52
ΔG_{SASA}	-27.43±4.41	-29.80±14.27	-21.94±2.76	-42.94±1.70	-55.017±1.68	-45.17±1.00	-72.17±5.24	-78.17±7.32
ΔG_{BIND}	**-60.88±1.67**	**-60.54±0.60**	**-39.94±2.71**	**-41.91±6.34**	**-57.13±3.15**	**-38.56±2.49**	**-81.58±2.44**	**-54.78±3.21**

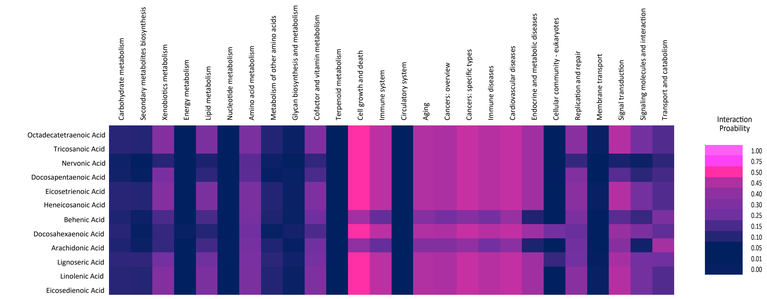


Fig 3: Metabolic Network Simulation of top performing fish fatty acid molecules. Heatmap diagram showing interaction proabbility of molecules with atherosclerosis associated KEGG metabolic pathways

Discussion

Pharmacokinetic properties and toxicity predictions reveal 23 of the 32 fatty acids to have optimum bioavailability and safety metrics. All studied compounds were predicted to have drug-like properties according to established rules; nevertheless, we excluded 3 probable carcinogenic lipids from further screening.

Docking simulations reveal multiple fatty acids to have potential therapeutic activities in different metabolic pathways.The protein-fatty acid complex formations were dominated by multiple short-length hydrophobic interactions in all the receptor binding-cavities. Binding affinities were significantly stronger than control drug molecules in case of pro-inflammatory receptors, as confirmed by molecular mechanics calculations. Nervonic acid, docosahexaenoic acid and docosapentaenoic acid were predicted to be the most active multiple pathway enzyme regulatory candidates.

Finally, metabolic network simulations reveal top docked fish fatty acids to have multi-dimensional interactions in atherosclerosis associated pathways; predominantly interacting with apoptotic, ageing, cancer, immunoregulatory and cardiovascular disease networks. Nervonic acid exhibited the highest overall interaction probability, with maximum interaction index to cell death regulation, cancer and cardiovascular disease pathways.

Table 2: Dietary fish sources of multiple top performing bioactive fatty acids

Fish Source	Predicted Atheroprotective Fatty Acids	Lipid Content (% dry weight)
Tuna	Nervonic acid, Docosahexaenoic acid, Eicosenoic acid, Tricosanoic acid, Linonelic acid, Behenic acid	6.3-18.4%
Fresh Water Eel	Nervonic acid, Docosahexaenoic acid, Tricosanoic acid, Linonelic acid, Behenic acid	3.6-20.4%
Lemuru	Nervonic acid, Docosahexaenoic acid, Tricosanoic acid, Linonelic acid, Behenic acid	2.0-2.5%
Cod	Docosahexaenoic acid, Linonelic acid, Docosapentaenoic acid	33.8-35.9%
Catfish	Docosahexaenoic acid, Octadodecatetraenoic acid, Linonelic acid,	13.0%
Mackerel	Docosahexaenoic acid, Linonelic acid	3.8%
Salmon	Docosahexaenoic acid, Linonelic acid, Docosapentaenoic acid	20.3-22.6%
Sardine	Docosahexaenoic acid, Linonelic acid, Docosapentaenoic acid	3.5-4.4%
Shoul Fish	Linonelic acid, Behenic acid	8.50%

Conclusion

- The present investigation reveals how a plethora of dietary fish fatty acids can ameliorate atherosclerosis associated pathophysiologies by multi-pathway regulation.
- Computational studies estimate safety, availability and potency of fish derived Nervonic acid and Docosahexanoic acid to be substantially superior to existing therapeutic standards.

Further Reading

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